

Model Predictive Control

Final exam 2020 solution proposal

Problem 1

Tasks:

- a) (15%) Considering N as the prediction horizon length, write the formulation for a model predictive controller. Your formulation should include the objective function and the constraints. Also clearly state the variables that you are going to optimize.

Suggested Solution:

Profit = Income – expenses. Income is the money (\$/sec) earned by selling the total oil produced from the three wells. Expense is the total cost associated in compressing the gas to be injected into the three wells (also in \$/sec).

The variables that should be optimized is the gas injection flow rates for each oil well i.e. $w_{g,i}$ for $i = 1, 2, 3$. The objective function can be written in various forms (all of which are accepted answers).

Here, λ_1, λ_2 are the scaling factors and Q_k, P_k are the weighting matrices. All other symbols are already

$\min_{(w_{g,i})}$	$J = \sum_{k=1}^N -C_{oil} \left(\sum_{i=1}^3 w_{o,i} \right)^T Q_k \left(\sum_{i=1}^3 w_{o,i} \right) + \sum_{k=1}^N C_{gas} \left(\sum_{i=1}^3 w_{g,i} \right)^T P_k \left(\sum_{i=1}^3 w_{g,i} \right)$ ‘OR’ $\min_{(w_{g,i})} J = \lambda_1 \sum_{k=1}^N -C_{oil} \sum_{i=1}^3 w_{o,i}^k + \lambda_2 \sum_{k=1}^N C_{gas} \sum_{i=1}^3 w_{g,i}^k$	(1)
s. t	$w_{o,i}^{k+1} = f(w_{o,i}^k, w_{g,i}^k, \theta_i)$	(3)
	$\sum_{i=1}^3 w_{g,i}^k \leq w_{g,max}$	(4)
	$g(w_{g,i}^k, \theta_i) \leq P_{s,max}$	(5)

explained in the question.

- b) (10%) Assume that for a given value of $P_{s,max}$, the optimal control problem is infeasible. How will you handle it? What will be your assumption about the nature of the constraint associated with the variable P_s . Write the formulation for the optimal control problem with necessary adjustments in the objective and the constraints.

Suggested solution

If an optimal control problem is infeasible, it can be handled by relaxing the problem. Inequality constraint(s) are relaxed by using slack variable(s). However, not all inequality constraints can be relaxed. Only soft constraints can be relaxed to free the optimization problem from infeasibility. Since the optimal control problem is infeasible for a given value of $P_{s,max}$, we assume that the variable P_s is a soft constraint. To soften the inequality constraint, remember to minimize the amount of violation.

The formulation for the optimal control problem with necessary adjustments in the objective and the constraints is shown below.

$\min_{(w_{g,i}, S_H)} J_{mod} = \sum_{k=1}^N -C_{oil} \left(\sum_{i=1}^3 w_{o,i}^k \right)^T Q_k \left(\sum_{i=1}^3 w_{o,i}^k \right) + \sum_{k=1}^N C_{gas} \left(\sum_{i=1}^3 w_{g,i}^k \right)^T P_k \left(\sum_{i=1}^3 w_{g,i}^k \right) + \sum_{k=1}^N (S_H^k)^T S_1 S_H^k \quad (6)$ s. t $w_{o,i}^{k+1} = f(w_{o,i}^k, w_{g,i}^k, \theta_i) \quad (7)$ $\sum_{i=1}^3 w_{g,i}^k \leq w_{g,max} \quad (8a)$ $g(w_{g,i}^k, \theta_i) \leq P_{s,max} + S_H^k \quad (8b)$	(7) (8a) (8b)
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Here, J_{mod} is the modified objective function, $S_H^k \geq 0$ are the slack variables that are added to the optimization problem and $S_1 \geq 0$ is the weighting matrix for the slack variables.

Problem 2

Tasks:

(a) (8%) Find the expression for matrix H and vector c .

Suggested solution:

The equivalent linear model of the process in discrete time as,

$$x_{k+1} = Ax_k + Bu_k + \Delta t \left(f(x_{op}, u_{op}, \theta) - (A_c x_{op} + B_c u_{op}) \right) \quad x_{k=0} \text{ is known,} \quad (9)$$

$$y_k = Cx_k + g(x_{op}, \theta) - C_c x_{op}. \quad (10)$$

After the operating points for the states, control inputs and outputs have been chosen or calculated, they remain constant. For a given value of Δt the expression $\Delta t \left(f(x_{op}, u_{op}, \theta) - (A_c x_{op} + B_c u_{op}) \right)$ from the state equation remains constant. Similarly the expression $g(x_{op}, \theta) - C_c x_{op}$ from the measurement equation remains constant. The linear model can then be expressed as,

$$x_{k+1} = Ax_k + Bu_k + v \quad x_{k=0} \text{ is known,} \quad (11)$$

$$y_k = Cx_k + w, \quad (12)$$

where $v = \Delta t \left(f(x_{op}, u_{op}, \theta) - (A_c x_{op} + B_c u_{op}) \right)$ and $w = g(x_{op}, \theta) - C_c x_{op}$.

The LQ optimal control problem taken into consideration is,

$$J = \frac{1}{2} \sum_{k=1}^N e_k^T P_k e_k + u_{k-1}^T Q_{k-1} u_{k-1} + \Delta u_{k-1}^T R_{k-1} \Delta u_{k-1}, \quad (13)$$

$$J = \frac{1}{2} (e_1^T P_1 e_1 + \dots, e_N^T P_N e_N + u_0^T Q_0 u_0 + \dots, u_{N-1}^T Q_{N-1} u_{N-1} + \Delta u_0^T R_0 \Delta u_0 + \dots, \Delta u_{N-1}^T R_{N-1} \Delta u_{N-1}) \quad (14)$$

s. t,

$$x_{k+1} = Ax_k + Bu_k + v, \quad x_0 \text{ is known,} \quad (15)$$

$$y_k = Cx_k + w, \quad (16)$$

$$e_k = r_k - y_k \quad (17)$$

$$\Delta u_k = u_k - u_{k-1} \quad (18)$$

We want to express equations (13) – (18) as a standard QP problem of the form,

$$\min_Z \quad \frac{1}{2} Z^T H Z + c^T Z, \quad (19)$$

s. t

$$A_\varepsilon z = b_\varepsilon, \quad (20)$$

$$A_i z \leq b_i, \quad (21)$$

$$z_L \leq z \leq z_H. \quad (22)$$

The vector of unknowns, z , is defined as,

$$z^T = [u^T, \Delta u^T, e^T, y^T, x^T] \quad (23)$$

Where,

$$\begin{aligned} u^T &= [u_0^T, \dots, u_{N-1}^T] \\ \Delta u^T &= [\Delta u_0^T, \dots, \Delta u_{N-1}^T] \\ e^T &= [e_1^T, \dots, e_N^T] \\ y^T &= [y_1^T, \dots, y_N^T] \\ x^T &= [x_1^T, \dots, x_N^T] \end{aligned} \quad (24)$$

The number of unknowns are $n_z = N (n_u + n_u + n_y + n_y + n_x)$.

Now $\frac{1}{2} z^T H z$ from equation (18) can be written as,

$$\frac{1}{2} z^T H z = \frac{1}{2} \begin{pmatrix} u \\ \Delta u \\ e \\ y \\ x \end{pmatrix}^T \begin{pmatrix} H_{11} & 0 & 0 & 0 & 0 \\ 0 & H_{22} & 0 & 0 & 0 \\ 0 & 0 & H_{33} & 0 & 0 \\ 0 & 0 & 0 & H_{44} & 0 \\ 0 & 0 & 0 & 0 & H_{55} \end{pmatrix} \begin{pmatrix} u \\ \Delta u \\ e \\ y \\ x \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{pmatrix}^T \begin{pmatrix} u \\ \Delta u \\ e \\ y \\ x \end{pmatrix} \quad (25)$$

$$\frac{1}{2} z^T H z = \frac{1}{2} (u^T H_{11} u + \Delta u^T H_{22} \Delta u + e^T H_{33} e + y^T H_{44} y + x^T H_{55} x) \quad (26)$$

Comparing (26) and (14) we get,

$$u^T H_{11} u = \begin{pmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{pmatrix}^T \begin{bmatrix} Q_0 & 0 & \dots & 0 \\ 0 & Q_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & Q_{N-1} \end{bmatrix} \begin{pmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{pmatrix} \quad (27)$$

If $Q_0 = Q_1 = \dots = Q_{N-1} = Q$, then we have,

$H_{11} = \begin{bmatrix} Q & 0 & \dots & 0 \\ 0 & Q & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & Q \end{bmatrix} = I_N \otimes Q$	(28)
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where $\otimes$ is the Kronecker product between matrices. In addition, comparing (26) and (14) we get,

$$\Delta u^T H_{22} \Delta u = \begin{pmatrix} \Delta u_0 \\ \Delta u_1 \\ \vdots \\ \Delta u_{N-1} \end{pmatrix}^T \begin{bmatrix} R_0 & 0 & \dots & 0 \\ 0 & R_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & R_{N-1} \end{bmatrix} \begin{pmatrix} \Delta u_0 \\ \Delta u_1 \\ \vdots \\ \Delta u_{N-1} \end{pmatrix} \quad (29)$$

If $R_0 = R_1 = \dots = R_{N-1} = R$, then we have,

$H_{22} = \begin{bmatrix} R & 0 & \dots & 0 \\ 0 & R & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & R \end{bmatrix} = I_N \otimes R$	(28)
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Also,

$$e^T H_{33} e = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{pmatrix}^T \begin{bmatrix} P_1 & 0 & \dots & 0 \\ 0 & P_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & P_N \end{bmatrix} \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_N \end{pmatrix} \quad (27)$$

If $P_1 = P_2 = \dots = P_N = P$, then we have,

$H_{33} = \begin{bmatrix} P & 0 & \dots & 0 \\ 0 & P & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & P \end{bmatrix} = I_N \otimes P$	(28)
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There are no 'y' and 'x' terms in equation (14), so we have,

$H_{44} = I_N \otimes 0_{n_y} = 0_{N.n_y}$	(29)
$H_{55} = I_N \otimes 0_{n_x} = 0_{N.n_x}$	(30)

Then we have,

$H = blkdiag(H_{11}, H_{22}, H_{33}, H_{44}, H_{55})$	(31)
$H = blkdiag(I_N \otimes Q, I_N \otimes R, I_N \otimes P, 0_{N.n_y}, 0_{N.n_x})$	

Here *blkdiag* means block diagonal matrix.

Also, in equation (14) we do not have linear terms, so we have,

$c = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{pmatrix} = 0_{n_z \times 1}$	(32)
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(b) (15%) Find the expression for matrix A_ϵ and vector b_ϵ .

Suggested Solution:

Now let us express the equality constraints of equations 15-18 in the form $A_\epsilon z = b_\epsilon$. We have in total 4 equality constraints, therefore, let us group them into 4 groups as,

$$\begin{bmatrix} A_{\epsilon,1u} & A_{\epsilon,1\Delta u} & A_{\epsilon,1e} & A_{\epsilon,1y} & A_{\epsilon,1x} \\ A_{\epsilon,2u} & A_{\epsilon,2\Delta u} & A_{\epsilon,2e} & A_{\epsilon,2y} & A_{\epsilon,2x} \\ A_{\epsilon,3u} & A_{\epsilon,3\Delta u} & A_{\epsilon,3e} & A_{\epsilon,3y} & A_{\epsilon,3x} \\ A_{\epsilon,4u} & A_{\epsilon,4\Delta u} & A_{\epsilon,4e} & A_{\epsilon,4y} & A_{\epsilon,4x} \end{bmatrix} \begin{bmatrix} u \\ \Delta u \\ e \\ y \\ x \end{bmatrix} = \begin{bmatrix} b_{\epsilon,1} \\ b_{\epsilon,2} \\ b_{\epsilon,3} \\ b_{\epsilon,4} \end{bmatrix} \quad (33)$$

Each row of equation (33) corresponds to each equality constraints of equations 15-18.

Let us first consider the equality constraints given by equation (15),

$$x_{k+1} = Ax_k + Bu_k + v \quad \rightarrow \quad x_k - Ax_{k-1} - Bu_{k-1} = v \quad (34)$$

For, $k=1$ to $k=N$, we have,

$$\begin{aligned} x_1 - Bu_0 &= Ax_0 + v \\ x_2 - Ax_1 - Bu_1 &= v \\ &\vdots \\ x_N - Ax_{N-1} - Bu_{N-1} &= v \end{aligned} \quad (35)$$

Arranging equation (35) in matrix form we get,

$$\begin{bmatrix} \underbrace{\begin{bmatrix} -B & 0 & 0 & \dots & 0 \\ 0 & -B & 0 & \dots & 0 \\ 0 & 0 & -B & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & -B \end{bmatrix}}_{A_{\epsilon,1u}} & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{\epsilon,1\Delta u}} & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{\epsilon,1e}} & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{\epsilon,1y}} & \underbrace{\begin{bmatrix} I & 0 & 0 & \dots \\ -A & I & 0 & \dots \\ 0 & -A & I & \dots \\ \vdots & \vdots & \vdots & \ddots \\ 0 & 0 & \dots & -A \end{bmatrix}}_{A_{\epsilon,1x}} \end{bmatrix} \begin{bmatrix} u \\ \Delta u \\ e \\ y \\ x \end{bmatrix} = \begin{bmatrix} v \\ v \\ v \\ v \\ v \end{bmatrix}$$

$$\begin{bmatrix} u_0 \\ \vdots \\ u_{N-1} \\ \Delta u_0 \\ \vdots \\ \Delta u_{N-1} \\ e_1 \\ \vdots \\ e_N \\ y_1 \\ \vdots \\ y_N \\ x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} Ax_0 + v \\ v \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \underbrace{v}_{b_{\epsilon,1}} \end{bmatrix}$$

Therefor we have,

$A_{\epsilon,1u} = \begin{bmatrix} -B & 0 & \dots & 0 \\ 0 & -B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & -B \end{bmatrix} = -I_N \otimes B$	(36)
$A_{\epsilon,1\Delta u} = 0_{N.n_x \times N.n_u}$	(37)
$A_{\epsilon,1e} = 0_{N.n_x \times N.n_y}$	(38)
$A_{\epsilon,1y} = 0_{N.n_x \times N.n_y}$	(39)
$A_{\epsilon,1x} = \begin{bmatrix} I & 0 & 0 & \dots & 0 \\ -A & I & 0 & \dots & 0 \\ 0 & -A & I & \dots & 0 \\ \vdots & & & \ddots & \\ 0 & \dots & \dots & -A & I \end{bmatrix} = I_{N.n_x} - (I_{N,-1} \otimes A)$	(40)

Furthermore,

$b_{\epsilon,1} = \begin{bmatrix} Ax_0 + v \\ v \\ \vdots \\ v \end{bmatrix}$	(41)
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Now let us consider the equality constraint of equation (16).

$$y_k = Cx_k + w \quad \rightarrow \quad y_k - Cx_k = w \quad (42)$$

For, $k=1$ to $k=N$, we have,

$$\begin{aligned} y_1 - Cx_1 &= w \\ y_1 - Cx_2 &= w \\ \vdots & \quad \quad \quad \vdots \\ y_N - Cx_N &= w \end{aligned} \quad (43)$$

Arranging equation (43) in matrix form we get,

$$\begin{bmatrix} \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{\epsilon,2u}} & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{\epsilon,2\Delta u}} & \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}}_{A_{\epsilon,2e}} & \underbrace{\begin{bmatrix} I & 0 & 0 & \dots & 0 \\ 0 & I & 0 & \dots & 0 \\ 0 & 0 & I & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & I \end{bmatrix}}_{A_{\epsilon,2y}} & \underbrace{\begin{bmatrix} -C & 0 & 0 & \dots & 0 \\ 0 & -C & 0 & \dots & 0 \\ 0 & 0 & -C & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & -C \end{bmatrix}}_{A_{\epsilon,2x}} \end{bmatrix}$$

Therefor we have,

$$A_{\epsilon, 2u} = 0_{N.n_y \times N.n_u} \quad (44)$$

$$A_{\in, 2\Delta u} = 0_{N.n_y \times N.n_u}$$

$$A_{\epsilon, 2e} = 0_{N.n_y \times N.n_y}$$

$$A_{\epsilon, 2y} = \begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{bmatrix} = I_N \otimes I_{n_y} = I_{N.n_y}$$

$$A_{\epsilon, 2x} = \begin{bmatrix} -C & 0 & \cdots & 0 \\ 0 & -C & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -C \end{bmatrix} = -I_N \otimes C \quad (47)$$

$$b_{\epsilon,2} = \begin{bmatrix} w \\ w \\ \vdots \\ w \end{bmatrix}$$

(49)

Similarly, considering the equality constraints of equation (17) we get,

$$e_k = r_k - y_k \quad \rightarrow \quad e_k + y_k = r_k \quad (50)$$

For, $k=1$ to $k=N$, we have,

$$\begin{aligned} e_1 + y_1 &= r_1 \\ e_2 + y_2 &= r_2 \\ \vdots &\quad \quad \quad \vdots \\ e_N + y_N &= r_N \end{aligned} \tag{51}$$

Arranging equation (51) in matrix form we get,

$$\begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots & 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots & 0 & 0 & 0 & \dots & \vdots \\ \vdots & \ddots & 0 & \ddots & \vdots & 0 & 0 & 0 & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 & 0 & 0 & \dots & \dots & 0 \end{bmatrix} \vdots \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \ddots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix} \vdots \begin{bmatrix} I & 0 & 0 & \dots & 0 \\ 0 & I & 0 & \dots & 0 \\ 0 & 0 & I & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & I \end{bmatrix} \vdots \begin{bmatrix} I & 0 & 0 & \dots & 0 \\ 0 & I & 0 & \dots & 0 \\ 0 & 0 & I & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & I \end{bmatrix} \vdots \begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & \vdots \\ 0 & 0 & 0 & \dots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \dots & \dots & 0 \end{bmatrix}$$

$$\begin{bmatrix} u_0 \\ \vdots \\ u_{N-1} \\ \Delta u_0 \\ \vdots \\ \Delta u_{N-1} \\ e_1 \\ \vdots \\ e_N \\ y_1 \\ \vdots \\ y_N \\ x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \underbrace{r_N}_{b_{\epsilon,3}} \end{bmatrix}$$

Therefor we have,

$$A_{\epsilon,3u} = 0_{N.n_y \times N.n_u} \quad (52)$$

$$A_{\epsilon,3\Delta u} = 0_{N.n_y \times N.n_u} \quad (53)$$

$$A_{\epsilon,3e} = \begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{bmatrix} = I_N \otimes I_{n_y} = I_{N.n_y} \quad (55)$$

$$A_{\epsilon,3y} = \begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{bmatrix} = I_N \otimes I_{n_y} = I_{N.n_y} \quad (56)$$

$$A_{\epsilon,3x} = 0_{N.n_y \times N.n_x}$$

$$b_{\epsilon,3} = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_N \end{bmatrix} \quad (57)$$

Finally from the equality constraint of equation (18) we get,

$$\Delta u_k = u_k - u_{k-1} \rightarrow \Delta u_{k-1} - u_{k-1} + u_{k-2} = 0 \quad (58)$$

For, $k=1$ to $k=N$, we have,

$$\begin{aligned} \Delta u_0 - u_0 &= -u_{-1} \\ \Delta u_1 - u_1 + u_0 &= 0 \\ \vdots & \\ \Delta u_{N-1} - u_{N-1} + u_{N-2} &= 0 \end{aligned} \quad (59)$$

Arranging equation (59) in matrix form we get,

$$\left[\underbrace{\begin{bmatrix} -I & 0 & 0 & \cdots & 0 \\ I & -I & 0 & \cdots & \vdots \\ 0 & I & 0 & \cdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I & -I \end{bmatrix}}_{A_{\epsilon,4u}} : \underbrace{\begin{bmatrix} I & 0 & 0 & \cdots & 0 \\ 0 & I & 0 & \cdots & 0 \\ 0 & 0 & I & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & I \end{bmatrix}}_{A_{\epsilon,4\Delta u}} : \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 0 \end{bmatrix}}_{A_{\epsilon,4e}} : \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 0 \end{bmatrix}}_{A_{\epsilon,4y}} : \underbrace{\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \cdots & 0 \end{bmatrix}}_{A_{\epsilon,4x}} \right]$$

$$\begin{bmatrix} u_0 \\ \vdots \\ u_{N-1} \\ \Delta u_0 \\ \vdots \\ \Delta u_{N-1} \\ e_1 \\ \vdots \\ e_N \\ y_1 \\ \vdots \\ y_N \\ x_1 \\ \vdots \\ x_N \end{bmatrix} = \underbrace{\begin{bmatrix} -u_{-1} \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{bmatrix}}_{b_{\epsilon,4}}$$

Therefor we have,

$$A_{\epsilon,4u} = \begin{bmatrix} -I & 0 & 0 & \dots & 0 \\ I & -I & 0 & \dots & 0 \\ 0 & I & -I & \dots & 0 \\ \vdots & & & \ddots & \\ 0 & \dots & \dots & I & -I \end{bmatrix} = -I_{N.n_u} + (I_{N,-1} \otimes I_{n_u}) \quad (60)$$

$$A_{\epsilon,4\Delta u} = \begin{bmatrix} I & 0 & \dots & 0 \\ 0 & I & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & I \end{bmatrix} = I_N \otimes I_{n_u} = I_{N.n_u} \quad (61)$$

$$A_{\epsilon,4e} = 0_{N.n_u \times N.n_y} \quad (62)$$

$$A_{\epsilon,4y} = 0_{N.n_u \times N.n_y} \quad (63)$$

$$A_{\epsilon,4x} = 0_{N.n_u \times N.n_x} \quad (64)$$

$$b_{\epsilon,4} = \begin{bmatrix} -u_{-1} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (65)$$

Summing it up,

$$A_{\epsilon} = \begin{bmatrix} -I_N \otimes B & 0_{N.n_x \times N.n_u} & 0_{N.n_x \times N.n_y} & 0_{N.n_x \times N.n_y} & I_{N.n_x} - (I_{N,-1} \otimes A) \\ 0_{N.n_y \times N.n_u} & 0_{N.n_y \times N.n_u} & 0_{N.n_y \times N.n_y} & I_{N.n_y} & -I_N \otimes C \\ 0_{N.n_y \times N.n_u} & 0_{N.n_y \times N.n_u} & I_{N.n_y} & I_{N.n_y} & 0_{N.n_y \times N.n_x} \\ -I_{N.n_u} + (I_{N,-1} \otimes I_{n_u}) & I_{N.n_u} & 0_{N.n_u \times N.n_y} & 0_{N.n_u \times N.n_y} & 0_{N.n_u \times N.n_x} \end{bmatrix} \quad (66)$$

$b_{\epsilon} = \begin{bmatrix} Ax_0 + v \\ v \\ \vdots \\ v \\ w \\ w \\ \vdots \\ w \\ r_1 \\ r_2 \\ \vdots \\ r_N \\ -u_{-1} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$	(67)
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Problem 3

Tasks:

- (a) (7%) Write down the structure of the output delay model in a proper state space form. Clearly illustrate and write down the system matrices of the output delay model. You should show the contents of the system matrices of the output delay model.

Suggested solution:

Figure 1 shows a simplified diagram of an output delay.

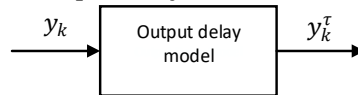


Figure 1: Output delay model

Without the output delay, the state space model of the process being controlled using MPC is given as,

$$x_{k+1} = Ax_k + Bu_k \quad (68)$$

$$y_k = Cx_k \quad (69)$$

Let us assume that the state space model from y_k to y_k^τ (i.e. the delay model) be written as,

$$x_{k+1}^\tau = A^\tau x_k^\tau + B^\tau y_k \quad (70)$$

$$y_k^\tau = C^\tau x_k^\tau \quad (71)$$

The number of samples corresponding to the computational time delay is given as,

$$n_\tau = \text{round down} \left(\frac{\tau}{\Delta t} \right)$$

$$n_\tau = \text{round down} \left(\frac{0.5}{0.1} \right) = 5$$

Illustration: For a computational delay of 5 samples i.e. $n_\tau = 5$,

$$\begin{bmatrix} x_{k+1}^1 \\ x_{k+1}^2 \\ x_{k+1}^3 \\ x_{k+1}^4 \\ x_{k+1}^5 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}}_{A^\tau} \underbrace{\begin{bmatrix} x_k^1 \\ x_k^2 \\ x_k^3 \\ x_k^4 \\ x_k^5 \end{bmatrix}}_{x_k^\tau} + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{B^\tau} y_k \quad (72)$$

$$y_k^\tau = \underbrace{[0 \ 0 \ 0 \ 0 \ 1]}_{C^\tau} \underbrace{\begin{bmatrix} x_k^1 \\ x_k^2 \\ x_k^3 \\ x_k^4 \\ x_k^5 \end{bmatrix}}_{x_k^\tau} \quad (73)$$

Analysing equations (72) and (73) we get,

$$\begin{aligned} x_{k+1}^1 &= y_k \\ x_{k+1}^2 &= x_k^1 = y_{k-1} \\ x_{k+1}^3 &= x_k^2 = y_{k-2} \\ x_{k+1}^4 &= x_k^3 = y_{k-3} \\ x_{k+1}^5 &= x_k^4 = y_{k-4} \end{aligned}$$

$$y_k^\tau = x_k^5 = y_{k-5}$$

i.e. there is an output delay of 5 samples.

(b) (5%) How would you use the output delay model together with the original state space model of the process in order to design a linear MPC?

Suggested solution:

To use the output delay model together with the original state space model of the process in order to design a linear MPC that can handle the computational time delay, the process model without output delay (equations 68 and 69) should be augmented with the output delay model (equations 70 and 71) as,

$$\underbrace{\begin{bmatrix} x_{k+1} \\ x_{k+1}^\tau \end{bmatrix}}_{\tilde{x}_{k+1}} = \underbrace{\begin{bmatrix} A & 0 \\ B^\tau C & A^\tau \end{bmatrix}}_{\tilde{A}} \underbrace{\begin{bmatrix} x_k \\ x_k^\tau \end{bmatrix}}_{\tilde{x}_k} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{\tilde{B}} u_k \quad (74)$$

$$y_k^\tau = \underbrace{[0 \ C^\tau]}_{\tilde{C}} \underbrace{\begin{bmatrix} x_k \\ x_k^\tau \end{bmatrix}}_{\tilde{x}_k} \quad (75)$$

Augmented model is in a standard form as,

$$\tilde{x}_{k+1} = \tilde{A}\tilde{x}_k + \tilde{B}u_k \quad (76)$$

$$y_k^\tau = \tilde{C}\tilde{x}_k \quad (77)$$

The linear MPC should be designed based on the augmented model of equations (76) and (77). The draw back with this idea is that additional states in x_k^τ are involved which makes the size of the control problem bigger than before.